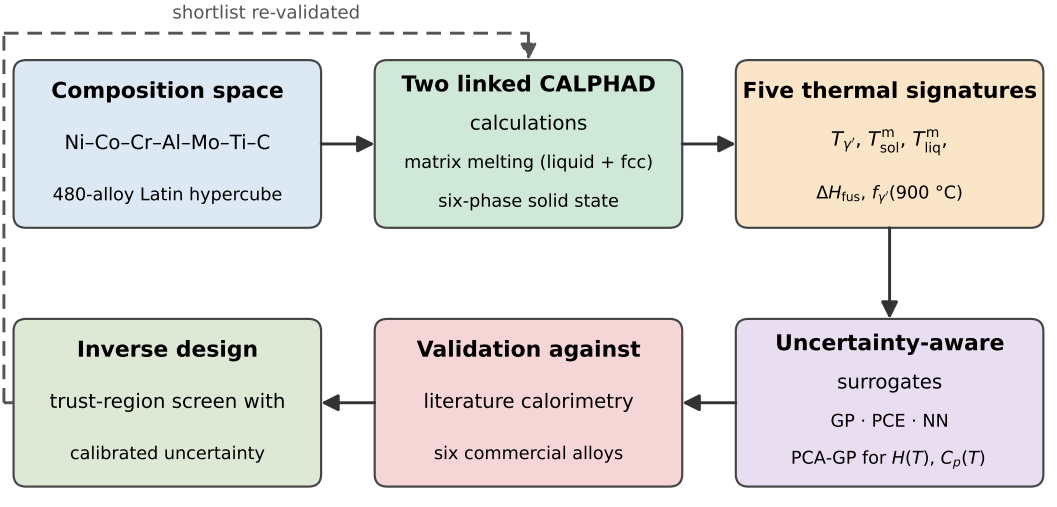
shortlist re-validated
Composition space
Ni–Co–Cr–Al–Mo–Ti–C
480-alloy Latin hypercube
Two linked CALPHAD
calculations
matrix melting (liquid + fcc)
six-phase solid state
Five thermal signatures
$T_{\gamma'}$, T^{m}_{sol}, T^{m}_{liq},
ΔH_{fus}, $f_{\gamma'}$(900 °C)
Uncertainty-aware
surrogates
GP · PCE · NN
PCA-GP for $H(T)$, $C_p(T)$
Validation against
literature calorimetry
six commercial alloys
Inverse design
trust-region screen with
calibrated uncertainty